US012411787B2

(12) **United States Patent**
Sarwar et al.

(10) **Patent No.: US 12,411,787 B2**
(45) **Date of Patent: Sep. 9, 2025**

(54) **SOFT DISCONNECT AND RECONNECT FOR USB TUNNELED PATH**

(71) Applicant: **Advanced Micro Devices, Inc.**, Santa Clara, CA (US)

(72) Inventors: **Zeeshan Sarwar**, Folsom, CA (US); **Preetesh Mahendra**, Folsom, CA (US)

(73) Assignee: **ADVANCED MICRO DEVICES, INC.**, Santa Clara, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 149 days.

(21) Appl. No.: **18/502,455**

(22) Filed: **Nov. 6, 2023**

(65) **Prior Publication Data**

US 2025/0147911 A1 May 8, 2025

(51) **Int. Cl.**
G06F 13/38 (2006.01)
G06F 13/42 (2006.01)

(52) **U.S. Cl.**
CPC ***G06F 13/387*** (2013.01); ***G06F 13/4282*** (2013.01)

(58) **Field of Classification Search**
CPC G06F 13/387; G06F 13/4282
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,387,348 B2 *	8/2019	Harriman	....	G06F 13/387
12,228,992 B2 *	2/2025	Regupathy	....	G06F 1/3228
2021/0365399 A1 *	11/2021	Dai	....	G06F 13/4221
2023/0289312 A1 *	9/2023	Rogers	....	G06F 13/4068

OTHER PUBLICATIONS

Welcome to the 80Gpbs Ultra-High Speed Era of USB4 [online]; Granite River Labs, GRL Team, Jan. 17, 2023 [retreived May 23, 2025], Retrieved from the Internet <URL: https://www.graniteriverlabs.com/en-us/technical-blog/usb4-80-cio80>. (Year: 2023).*

Kumar, Sanjeet; USB3 Gen T Tunneling Over USB4 [online]; Jan. 16, 2023 [retreived May 22, 2025]; Retreived from the Internets <URL: https://community.cadence.com/cadence_blogs_8/b/fv/posts/usb3-gen-t-tunneling-over-usb4>. (Year: 2023).*

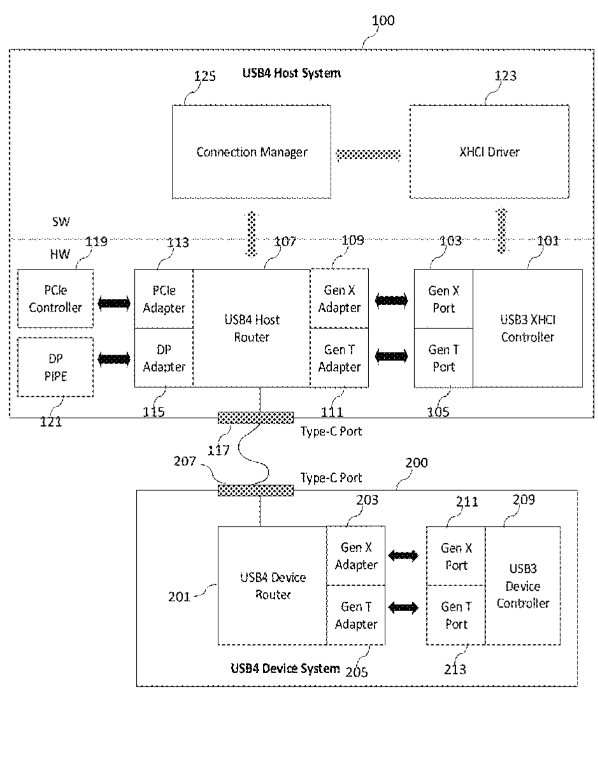
* cited by examiner

Primary Examiner — Eric T Oberly
(74) *Attorney, Agent, or Firm* — Slater Matsil, LLP

(57) **ABSTRACT**

A method of operating a Universal Serial Bus (USB) host system includes: disabling, by a USB host driver of the USB host system, a first tunneled path between a first port of a USB host controller of the USB host system and a first port of a USB device controller of a USB device, where the first tunneled path is configured to, through protocol tunneling, transmit first data packets of a first protocol format; sending, by the USB host driver, a request for protocol switching to a connection manager of the USB host system; disconnecting, by the connection manager, the first tunneled path; and establishing, by the connection manager, a second tunneled path between a second port of the USB host controller and a second port of the USB device controller.

20 Claims, 7 Drawing Sheets



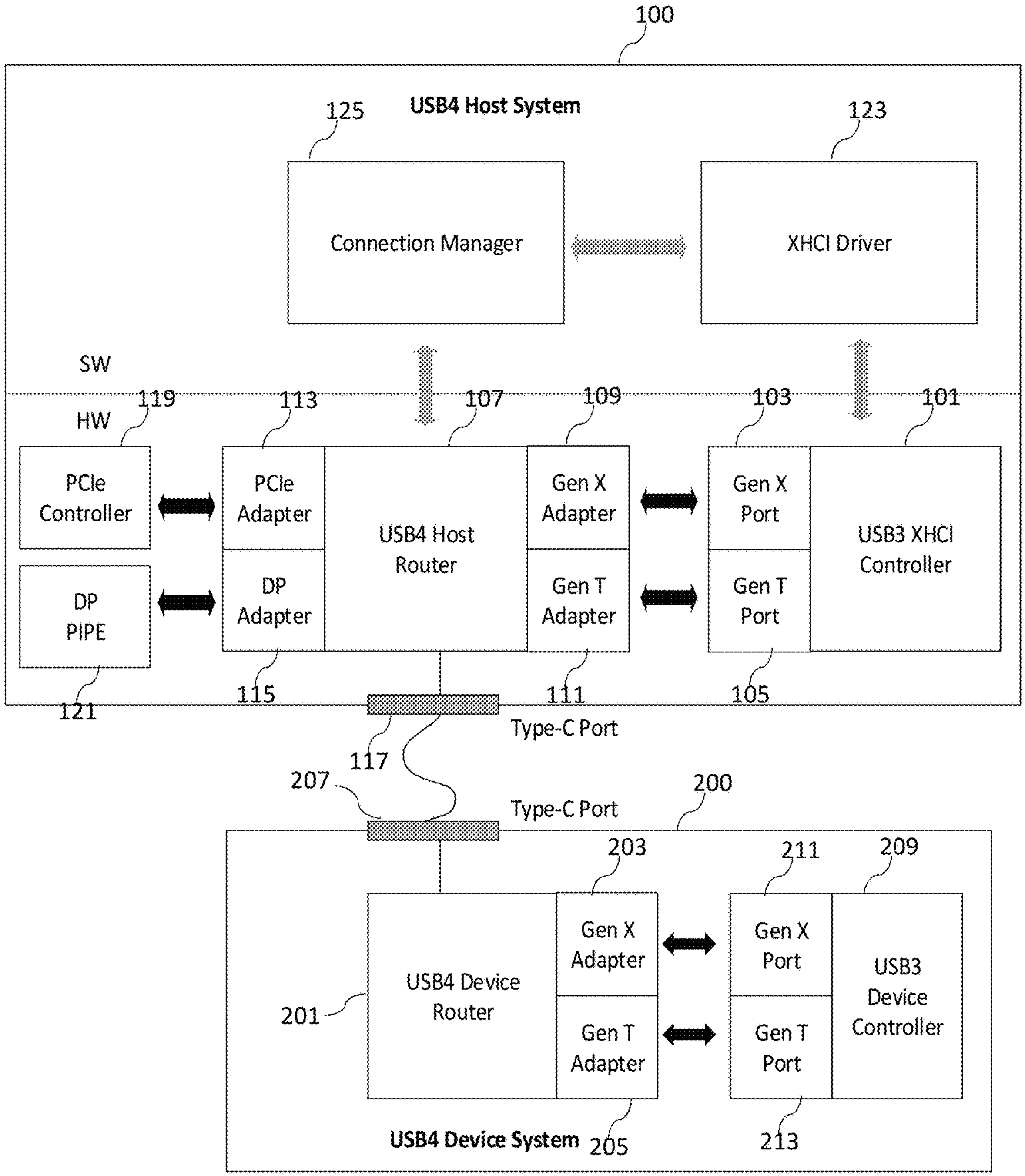

100
USB4 Host System
125
Connection Manager
123
XHCI Driver
SW
HW
119
113
107
109
103
101
PCIe Controller
PCIe Adapter
USB4 Host Router
Gen X Adapter
Gen X Port
USB3 XHCI Controller
DP PIPE
DP Adapter
Gen T Adapter
Gen T Port
121
115
111
105
Type-C Port
117
207
Type-C Port
200
203
211
209
Gen X Adapter
Gen X Port
201
USB4 Device Router
USB3 Device Controller
Gen T Adapter
Gen T Port
USB4 Device System
205
213

Fig. 1

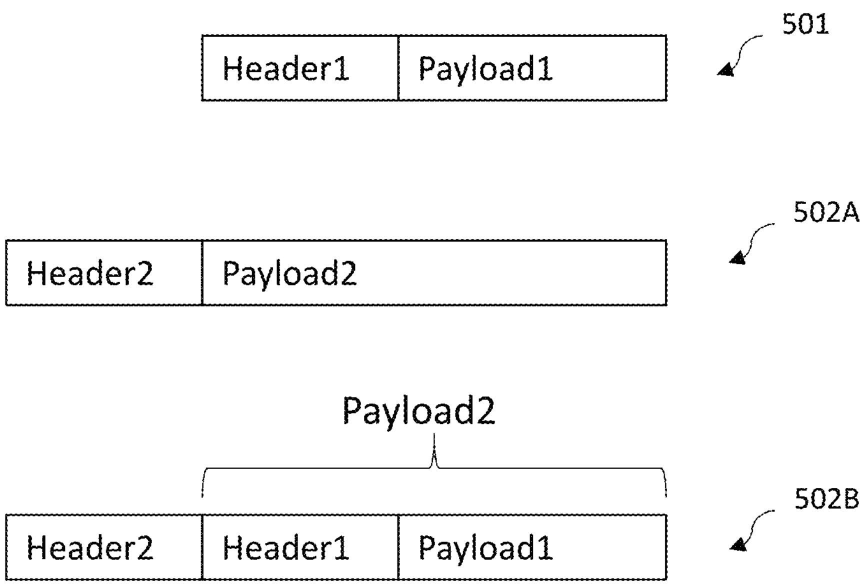
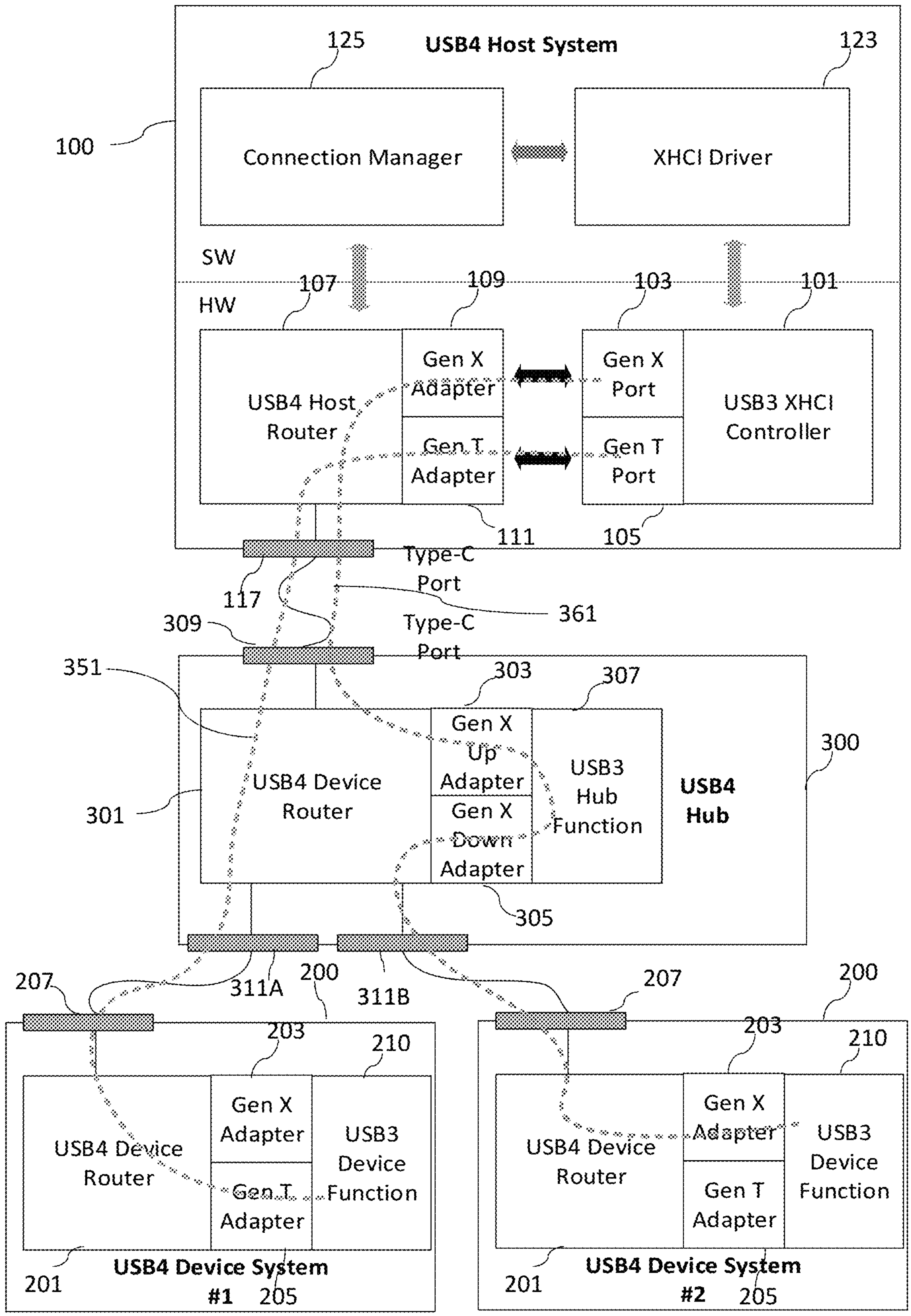
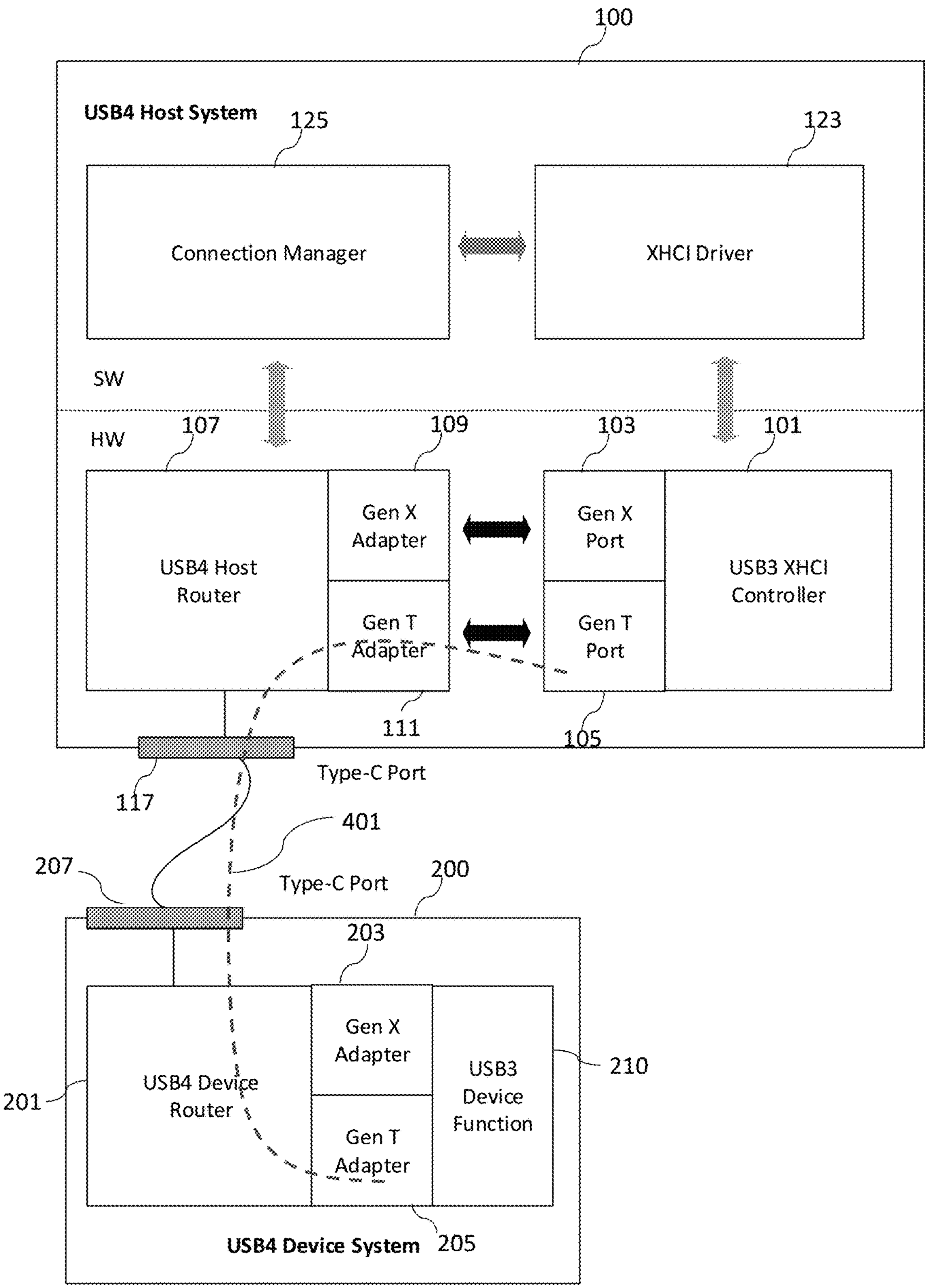

501
Header1
Payload1
502A
Header2
Payload2
Payload2
502B
Header2
Header1
Payload1

Fig. 2



USB4 Host System
125
123
100
Connection Manager
XHCI Driver
SW
HW
107
109
103
101
USB4 Host Router
Gen X Adapter
Gen T Adapter
Gen X Port
Gen T Port
USB3 XHCI Controller
111
105
Type-C Port
117
361
309
Type-C Port
351
303
307
300
301
USB4 Device Router
Gen X Up Adapter
Gen X Down Adapter
USB3 Hub Function
USB4 Hub
305
207
311A
200
311B
207
200
203
210
203
210
USB4 Device Router
Gen X Adapter
Gen T Adapter
USB3 Device Function
USB4 Device Router
Gen X Adapter
Gen T Adapter
USB3 Device Function
201
USB4 Device System #1
205
201
USB4 Device System #2
205

Fig. 3



100
USB4 Host System
125
123
Connection Manager
XHCI Driver
SW
HW
107
109
103
101
USB4 Host Router
Gen X Adapter
Gen T Adapter
Gen X Port
Gen T Port
USB3 XHCI Controller
111
105
Type-C Port
117
401
207
Type-C Port
200
203
USB4 Device Router
Gen X Adapter
Gen T Adapter
USB3 Device Function
210
201
205
USB4 Device System

Fig. 4

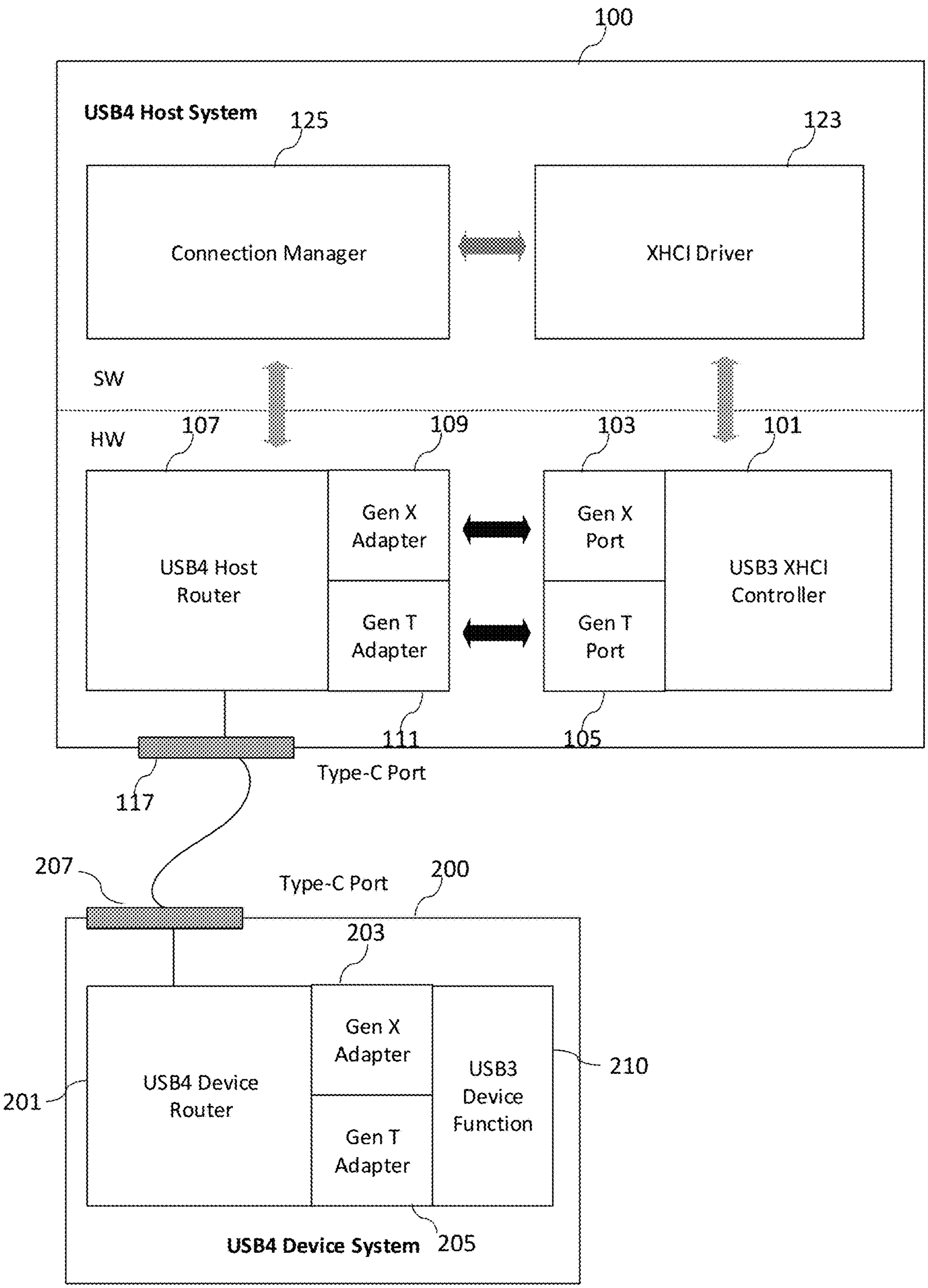

100
USB4 Host System
125
123
Connection Manager
XHCI Driver
SW
HW
107
109
103
101
USB4 Host Router
Gen X Adapter
Gen T Adapter
Gen X Port
Gen T Port
USB3 XHCI Controller
111
105
Type-C Port
117
207
Type-C Port
200
203
USB4 Device Router
Gen X Adapter
Gen T Adapter
USB3 Device Function
210
201
USB4 Device System
205

Fig. 5

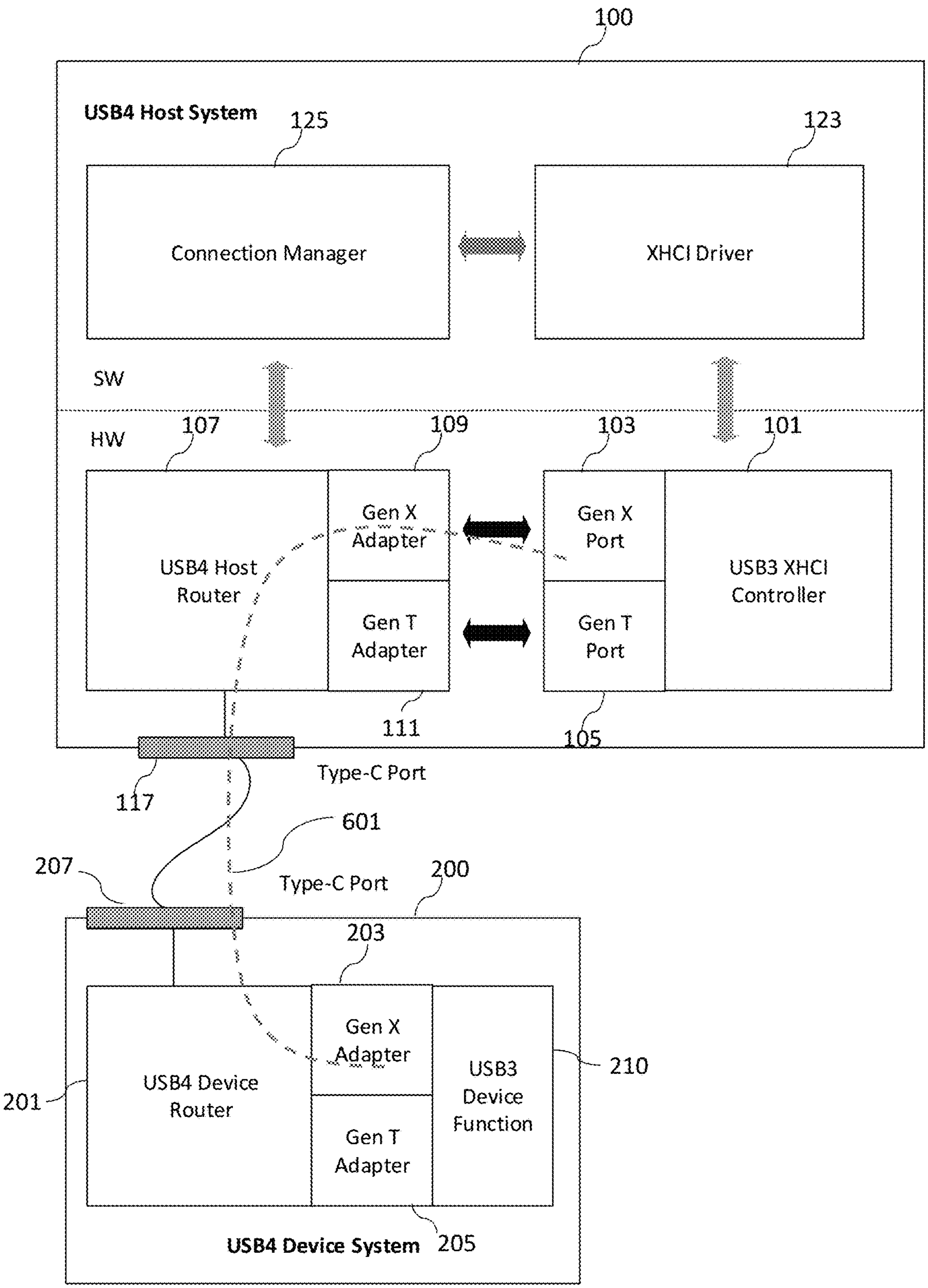

100
USB4 Host System
125
123
Connection Manager
XHCI Driver
SW
HW
107
109
103
101
Gen X
Adapter
Gen X
Port
USB4 Host
Router
USB3 XHCI
Controller
Gen T
Adapter
Gen T
Port
111
105
Type-C Port
117
601
207
Type-C Port
200
203
Gen X
Adapter
USB3
Device
Function
210
201
USB4 Device
Router
Gen T
Adapter
USB4 Device System
205

Fig. 6

700

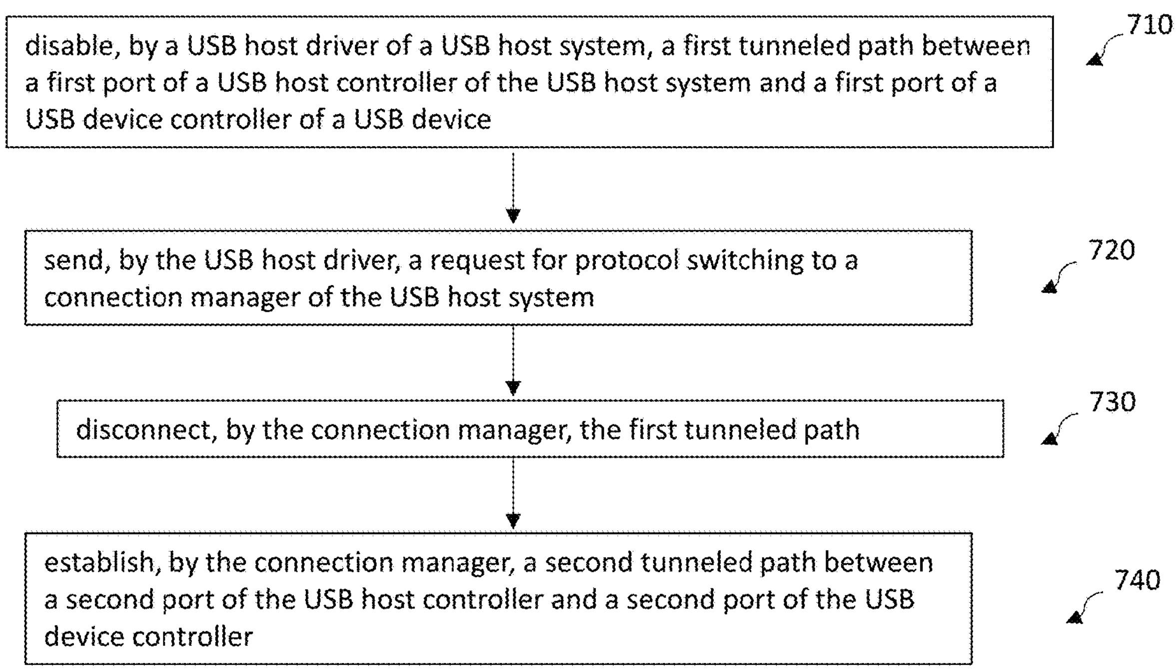

disable, by a USB host driver of a USB host system, a first tunneled path between a first port of a USB host controller of the USB host system and a first port of a USB device controller of a USB device
710
send, by the USB host driver, a request for protocol switching to a connection manager of the USB host system
720
disconnect, by the connection manager, the first tunneled path
730
establish, by the connection manager, a second tunneled path between a second port of the USB host controller and a second port of the USB device controller
740

Fig. 7

SOFT DISCONNECT AND RECONNECT FOR USB TUNNELED PATH

TECHNICAL FIELD

The present invention relates generally to Universal Serial Bus (USB) systems and the operation of USB systems.

BACKGROUND

Universal Serial Bus (USB) standard is an industry standard that specifies the physical interfaces and protocols for connecting, data transferring, and powering of hosts (e.g., personal computers), peripherals (e.g., keyboards and mobile devices), and intermediate hubs. USB was designed to standardize the connection of peripherals to computers, replacing various interfaces such as serial ports, parallel ports, and game ports. The USB connection has become commonplace on a wide range of devices, such as keyboards, mice, cameras, printers, scanners, flash drives, smartphones, game consoles, and power banks.

Multiple generations of USB specifications have been developed over the years. With each newer generation of the USB specification, the USB standard supports a higher data rate. Generally, newer generations of USB systems support multiple operation modes or data rates and are backward compatible with older generations of systems. Due to the multiple modes and data rates supported by different USB systems, interoperability between USB systems with different capabilities still poses a technical challenge. There is a need in the art to improve the interoperability of USB systems.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

FIG. **1** illustrates block diagrams of a USB host system and a USB device system in an implementation;

FIG. **2** illustrates data packets used in protocol tunneling in an implementation;

FIG. **3** illustrates connections between a USB host system and two USB device systems through a USB hub in an implementation;

FIGS. **4-6** illustrate a protocol switching flow for tunneled paths between a USB host system and a USB device system in an implementation; and

FIG. **7** illustrates a flow chart of a method of operating a USB host system in some implementations.

DETAILED DESCRIPTION OF ILLUSTRATIVE EXAMPLES

The making and using of the presently disclosed examples are discussed in detail below. It should be appreciated, however, that the present invention provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific examples discussed are merely illustrative of specific ways to make and use the invention, and do not limit the scope of the invention. Throughout the discussion herein, unless otherwise specified, the same or similar reference numerals or labels in different figures refer to the same or similar component or signal.

The present disclosure will be described with respect to examples in a specific context, namely soft disconnect and reconnect of tunneled paths between USB systems (e.g., between a USB host system and a USB device system) to achieve dynamic protocol switching. Note that while USB4 systems may be used as examples in the discussion herein, skilled artisans will readily appreciate that the principle disclosed herein is generic and may be applied to other USB systems, e.g., future generations of USB systems (e.g., USB systems based on future versions of USB specifications).

Currently, there are four generations of USB specifications: USB 1. “x”, USB 2.0, USB 3. “x”, and USB4. The USB4 specification enhances the data transfer and power supply functionality with connection-oriented, tunneling architecture designed to combine multiple protocols onto a single physical interface, so that the total speed and performance of the USB4 Fabric can be dynamically shared.

The USB4 version 2 specification defines an additional USB3 tunneling protocol: Gen T tunneling protocol. This protocol is in addition to the existing Gen X tunneling protocol which was defined in USB4 version 1. Gen T tunneling protocol supports full 80 Gbps USB4 version 2 bandwidth, while Gen X tunneling is based on the USB3.2 specification, and is limited to a maximum bandwidth of 20 Gbps.

FIG. **1** illustrates block diagrams of a USB host system **100** and a USB device system **200** in an implementation. The USB host system **100** (may also be referred to as a USB host) may be, e.g., a computer, and the USB device system **200** (may also be referred to as a USB device) may be, e.g., a USB drive, a digital camera, a mouse, or the like. Note that for simplicity, not all features of the USB host system **100** and the USB device system **200** are illustrated.

As illustrated in FIG. **1**, the USB host system **100** includes a USB host controller **101**, a USB host router **107**, a USB host driver **123**, and a connection manager **125**. The USB host controller **101** and the USB host router **107** are hardware and are shown in the hardware (HW) domain of the USB host system **100** in FIG. **1**. The USB host driver **123** and the connection manager **125** are software and are shown in the software (SW) domain of the USB host system **100** in FIG. **1**.

In the example of FIG. **1**, the USB host controller **101** is a USB3 extensible Host Controller Interface (XHCI) controller, and the USB host driver **123** is an XHCI driver. These are, of course, non-limiting examples. The USB host controller **101** and the USB host driver **123** may be any suitable USB host controller and any suitable USB host driver. The USB host controller **101** controls how the hardware of the USB host system **100** behaves, e.g., how the USB host system **100** accesses the physical channel (e.g., USB connectors **117**, such as Type-C connectors) and transmits (e.g., send or receive) electrical signals over the physical channel.

The USB host driver **123** is a software program (e.g., a device driver) that drives the USB host controller **101** and provides a software interface to the hardware of the USB host system **100**. The USB host driver **123** enables the operating system (OS) and other computer programs to access USB hardware functions without needing to know precise details about the hardware being used. The USB host driver **123** may run on, e.g., the processor of the computer and may be stored in, e.g., the Basic Input/Output System (BIOS) of the computer. USB host controller and USB host driver are defined in USB specifications and known in the art, and the details are not discussed here.

The connection manager **125** is a software program that functions to, among other things, establish path(s) (e.g., tunneled paths) between the USB host controller **101** and USB device controller **209** of USB device(s) connected to the USB host system **100**. Other functions of the connection manager **125** include: managing the protocol adapters (e.g., **109**, **111**, **113**, **115**) of the USB host router **107**, programming the protocol adapters, and bandwidth management (e.g., deciding the bandwidth (e.g., data rate) assigned for each tunneled protocol). The connection manager **125** may run on, e.g., the processor of the computer. USB connection manager is defined in USB specifications and known in the art, and the details are not discussed here.

In the example of FIG. **1**, the USB host router **107** (e.g., a USB4 host router) provides a high-speed (e.g., up to 80 Gbps) link channel (e.g., a data link layer channel) between the USB host router **107** and the USB device router **201** of the USB device **200** connected to the USB host system **100**. The link channel is shared by multiple controllers (e.g., **101**, **119**, **121**) of the USB host system **100** through protocol tunneling. The multiple controllers (e.g., **101**, **119**, **121**) may send data packets **501** (see in FIG. **2**) with different protocol formats (e.g., Gen X format, Gen T format, PCIe format, or Display Port (DP) format), and the data packets **501** with different protocol formats are encapsulated in data packets **502** (see FIG. **2**) transmitted between the USB host router **107** and the USB device router **201** of the connected USB device (e.g., a USB device supporting USB4 specification). In other words, the data packets **502** (e.g., data packets having protocol format defined by USB4 specification, also referred to as USB4 data packets) carry the data packets **501** with different protocol formats in the payload regions of the data packets **502**, thus allowing the data packets **501** with different protocol formats to be transmitted on the shared link channel as data packets **502**. This is referred to as protocol tunneling in the USB specification. FIG. **2** illustrates protocol tunneling.

FIG. **2** illustrates a data packet **501** and a data packet **502** in an implementation. In FIG. **2**, the data packet **502** is shown twice as **502**A and **502**B. The data packet **502**A shows the data packet **502** before the data packet **501** is packed (e.g., carried) in the payload region of the data packet **502**, and the data packet **502**B shows the data packet **502** after the data packet **501** is packed into the payload region of the data packet **502**.

In FIG. **2**, the data packet **501** may be, e.g., a data packet in Gen X protocol format, Gen T protocol format, PCIe protocol format, or DP protocol format. The data packet **501** has a header (labeled as "header**1**") and a payload (labeled as "payload**1**"). Similarly, the data packet **502** has a header (labeled as "header**2**") and a payload (labeled as "payload**2**"). Through protocol tunneling, the data packet **501** is packed (e.g., carried) in the payload region of the data packet **502**. The header of the data packet **502** may include bits to indicate the type (e.g., the protocol format) of the data packet **501** carried, the address (e.g., destination controller) of the carried data packet **501**, and so on. Once the data packet **502** reaches the destination system (e.g., **100** or **200**), the data packet **501** may be un-packed (e.g., recovered) from the payload region of the data packet **502** and sent to its intended recipient (e.g., **101**, **119**, or **121**).

Note that although USB4 packet and USB4 host router (and USB4 device router) are used in the illustrated implementations, these are merely non-limiting examples. A skilled artisan will readily appreciate that the principle disclosed herein can be easily applied to future generations of USB systems. For example, the USB host router **107** may be a future-generation host router, and the data packet **502** transmitted by the USB host router **107** may be in any future-generation data packet format beyond the USB4 data packet. These and other variations are fully intended to be included within the scope of the present disclosure.

Referring back to FIG. **1**, the USB host router **107** includes a plurality of protocol adapters, such as a Gen X adapter **109**, a Gen T adapter **111**, a Peripheral Component Interconnect Express (PCIe) adapter **113**, and a Display Port (DP) adapter **115**. The number and the type of adapters illustrated in FIG. **1** is a non-limiting example. Other numbers and other types of adapters are also possible and are fully intended to be included within the scope of the present disclosure. For example, the USB host router **107** may have one Gen X adapter **109** and multiple Gen T adapters **111**.

Each of the adapters of the USB host router **107** is used to pack and up-pack (see FIG. **2**) a data packet **501** of a corresponding protocol format for protocol tunneling. In other words, each adapter is used for protocol format conversion, e.g., converting between the data packet **501** and the data packet **502**.

FIG. **1** also illustrates a Gen X port **103** and a Gen T port **105** of the USB host controller **101**. The Gen X port **103** and the Gen T port **105** are used to transmit (e.g., send and/or receive) data packets in Gen X protocol format and Gen T protocol format, respectively. The Gen X port **103** is coupled to the Gen X adapter **109** of the USB host router **107**, and the Gen T port **105** is coupled to the Gen T adapter **111** of the USB host router **107**.

FIG. **1** further illustrates a PCIe controller **119** and a DP controller **121**. The PCIe adapter **113** of the USB host router **107** is coupled to the PCIe controller **119**. The DP adapter **115** of the USB host router **107** is coupled to the DP controller **121**.

As discussed above, the various controller (e.g., **101**, **119**, **121**) may send data packets **501** in different protocol formats. The data packets **501** are encapsulated as data packet **502** by the respective adapters (e.g., **109**, **111**, **113**, **115**) and sent through a USB connector **117** (e.g., a USB Type-C connector, also referred to as a USB Type-C port) to a connected USB device (or a USB hub).

The USB device system **200** in FIG. **1** includes a USB device controller **209** (e.g., a USB3 device controller) and a USB device router **201**. The USB device controller **209** includes a Gen X port **211** and a Gen T port **213** for transmitting data packets of Gen X protocol format and Gen T format, respectively. The USB device router **201** includes a Gen X adapter **203** coupled to the Gen X port **211** and includes a Gen T adapter **205** coupled to the Gen T port **213**. The functions of the Gen X adapter **203** and the Gen T adapter **205** are the same as or similar to the Gen X adapter **109** and the Gen T adapter **111**, respectively, and the details are not repeated here. The USB device router **201** communicates through a USB connector **207** (e.g., a USB Type-C connector) with the USB host router **107**. The USB device router **201** performs protocol tunneling for the USB device system **200**, similar to the USB host router **107**, and the details are not repeated here.

FIG. **3** illustrates connections between a USB host system **100** and two USB device systems **200** through a USB hub **300** in an implementation. The USB host system **100** and the USB device systems **200** are the same as or similar to those in FIG. **1**, thus details are not repeated here. Note that in FIG. **3** (and subsequent figures), for simplicity, the USB device controller **209** and the ports (e.g., Gen X port **211** and Gen

T port **213**) of the USB device controller **209** illustrated in FIG. **1** may be illustrated as one functional block referred to as USB device function **210**.

In FIG. **3**, the USB hub **300** includes a USB device router **301** (e.g., a USB4 device router), and a USB hub function **307** (e.g., USB3 hub function). The USB device router **301** includes a Gen X up adapter **303** and a Gen X down adapter **305**. In an implementation, the USB hub function **307** is a USB3 hub that performs the hub function in accordance with the USB3 specification. The USB hub **300** has a USB connector **309** (e.g., USB Type-C port) connected to the USB connector **117** of the USB host system **100**, and has two USB connectors **311**A and **311**B (e.g., USB Type-C ports) connected to the USB connectors **207** of two USB device systems **200**. The number of USB connectors shown in FIG. **3** is illustrative and non-limiting, and other numbers are also possible.

In some implementations, data packet **502** carrying Gen T data packets (e.g. data packet in Gen T protocol format) simply passes through the USB device router **301** of the USB hub **300**. For data packet **502** carrying Gen X data packets (e.g. data packet in Gen X protocol format), the USB device router **301** of the USB hub **300** routes the data packets **502** to the USB hub function **307**. In some implementations, the USB hub function **307** performs standard USB3 hub function (e.g., multiplexing data packets from multiple, different USB device systems **200**) and allows multiple USB device systems **200** with Gen X capability to communicate with one Gen X port **103** of the USB host controller **101**. In other words, the USB host controller **101** is able to transmit data packets **501** of Gen X protocol format to multiple USB devices **200** using only one Gen X port **103** and one Gen X adapter **109**. In contrast, the number of Gen T adapter **111** (or the number of Gen T port **105**) decides (e.g., equals) the number of USB devices **200** exchanging Gen T data packets with the USB host controller **101**.

In the example of FIG. **3**, the USB host router **107** of the USB host system **100** has one Gen T adapter **111**, and therefore, can only accommodate one USB device system **200** operating in Gen T mode (e.g., sending Gen T data packets). Both the USB device systems **200** in FIG. **3** support Gen T protocol and Gen X protocol. The first one of the USB device systems **200** is assigned a Gen T connection, and therefore, a tunneled path **351** is established between the Gen T port **105** of the USB host controller **101** and the Gen T port (not illustrated in FIG. **3**) of the USB device controller in the USB device function **210**. Since the USB host system **100** in the example of FIG. **3** only has one Gen T adapter **111**, the second one of the USB device systems **200** is assigned a Gen X connection, and therefore, a tunneled path **361** is established between the Gen X port **103** of the USB host controller **101** and the Gen X port (not illustrated in FIG. **3**) of the USB device controller in the USB device function **210**.

A Gen T device (e.g., a USB device supporting Gen T tunneling protocol) is required to be backward compatible to support Gen X tunneling. After discovering the device router capabilities, the connection manager's preference is to establish a Gen T path between the USB host (e.g., a USB3 host) and the USB device (e.g., a USB3 device) through protocol tunneling. If the USB device or the USB host does not support Gen T protocol, or there is no free Gen T port available in the USB host controller, the connection manager would set up a Gen X path between the USB host and the USB device.

The current USB specification (e.g., the USB4 specification) assumes that once a Gen T path or a Gen X path is established, it will remain intact until the USB device is physically disconnected. In other words, once a USB tunneled path is established through USB4 protocol tunneling for a particular protocol (Gen X or Gen T), the USB specification does not define any method to tear down (e.g., disconnect) the tunneled path and establish a new tunneled path for an alternate protocol, while the USB device remains physically connected to the USB host. This disclosure discloses new capabilities and flow (e.g., method) to achieve soft disconnect and reconnect of tunneled paths for USB systems.

There are many reasons or scenarios where the ability to dynamically (e.g., on-the-fly, or on-demand) switch the tunneling protocol while the USB device remains physically connected to the USB host is advantageous. Consider a scenario where the USB host architecture supports partial features of the USB specification, and the USB device expects transfer modes which the USB host does not support. For example, a particular USB host implementation may not support isochronous transfer for Gen T protocol tunneling. This implementation can make the USB host controller design much simpler without sacrificing much performance because isochronous transfer is an optional feature for Gen T, and it is unlikely that a USB device vendor will implement it. In the example discussed here, the USB device does support isochronous transfer, and the USB host driver discovers that the USB device wants to communicate through isochronous transfer mode only after the USB connection is established and the USB host driver parses the USB device descriptors. With the protocol switching flow (e.g., method to switch tunneling protocol) disclosed herein in place, the USB host driver can ask the connection manager to switch from Gen T connection (e.g., a tunneled path using Gen T protocol tunneling) to Gen X connection (e.g., a tunneled path using Gen X protocol tunneling) to enable isochronous transfer. Without the protocol switching flow disclosed herein, the USB host driver may have to report that the USB device is not supported.

As another example, when a USB device (e.g., a USB4 device that supports both Gen X and Gen T protocols) is plugged in, there may not be a Gen T downstream port available in the USB host controller. Therefore, the connection manager enables the USB device and establishes a Gen X connection. Later, some other Gen T device is disconnected from the USB host, freeing up a Gen T downstream port in the USB host controller. Noticing the available downstream Gen T port, the connection manager can initiate a Gen X soft disconnect (to disconnect the Gen X tunneled path) and reestablish a Gen T connection for the USB device. This allows the USB device to achieve higher throughput opportunistically (e.g., whenever possible).

As yet another example, there might be compatibility issues with certain USB devices operating in Gen T mode. With the protocol switching flow in place, the USB host driver can, after discovering an incompatibility issue of a connected USB device, request the connection manager to soft disconnect the Gen T path for the USB device and reestablish connection through a Gen X path. By downgrading the connection of the USB device to a legacy mode (e.g., Gen X mode), better compatibility is achieved when needed. An example of protocol switching flow is illustrated below with reference to FIGS. **4**-**6**.

In FIG. **4**, after the USB device **200** (e.g., a USB device that supports Gen X and Gen T mode) is plugged into the USB connector **117** of the USB host system **100**, a tunneled path **401** of Gen T type (e.g., using Gen T protocol tunneling) is initially established between the Gen T port **105** of the

USB host controller **101** of the USB host system **100** and the Gen T port **213** (not illustrated in FIG. **4** but illustrated in FIG. **1**) of the USB device controller **209** of the USB device **200**. The process of detecting a newly plugged-in USB device and assigning a connection type (e.g., Gen T or Gen X) for the USB device is described in USB specifications and known in the art, and the details are not discussed here.

Next, the USB host driver **123** receives descriptors of the USB device **200**. The descriptors may contain information such as the operation modes and/or data rates supported by the USB device **200**. The descriptors may be sent from the USB device **200** to the USB host **100** through a regular data channel between the USB device **200** and the USB host **100**, where the regular data channel is different from the tunneled path **401**. After parsing the descriptors, the USB host driver **123** finds out that the Gen T mode of the USB device **200** is not compatible with that of the USB host **100**, and therefore, the USB host driver **123** triggers a protocol switching flow in order to soft disconnect the Gen T path (e.g., **401**) and establish a Gen X path. Details are discussed below.

In some implementations, in a first step of the protocol switching flow, the USB host driver **123** disables the downstream port of the initial tunneled path **401**. In the illustrated example of FIG. **4**, the downstream port is the Gen T port **105**. The USB host driver **123** may disable the downstream port (e.g., **105**) by programming a control bit (e.g., writing a value of 0 for the control bit) for the downstream port in a control register of the USB host controller **101**. Disabling the downstream port breaks down (e.g., prohibits or stops) data transmission on the tunneled path **401**. Therefore, disabling the downstream port (e.g., **105**) may also be referred to as disabling the corresponding tunneled path (e.g., **401**).

Next, in a second step of the protocol switching flow, the USB host driver **123** send a protocol switching request to the connection manager **125**. For example, the USB host driver **123** may request to switch from the Gen T type connection to a Gen X type connection for the USB device **200**.

Referring next to FIG. **5**, in a third step of the protocol switching flow, the connection manager **125** tears down (e.g., disconnect) the tunneled path **401**, e.g., by disabling the Gen T adapter **111** of the USB host router **107** and the Gen T adapter **205** of the USB device router **201**. Since the tunneled path **401** is disconnected in the processing step of FIG. **5**, the tunneled path **401** shown in FIG. **4** is no longer shown in FIG. **5**. The connection manager **125** may disable the Gen T adapters **111** and **205** by programming the Path Enable (PE) control bit (e.g., writing a value of 0 for the control bit) for the Gen T adapter **111** in a control register of the USB host router **107**, and by programming the Path Enable (PE) control bit (e.g., writing a value of 0 for the control bit) for the Gen T adapter **205** in a control register of the USB device router **201**. Disabling the Gen T adapters **111** and **205** disconnects (e.g., clears, removes) the tunneled path **401**, and results in a disconnect event in the USB host driver software stack. This achieves the soft disconnect of the tunneled path **401**. After the tunneled path **401** is disconnected, the connection manager **125** informs the USB host driver **123** of the disconnection of the tunneled path **401**.

Referring next to FIG. **6**, in a third step of the protocol switching flow, the connection manager **125** establishes a tunneled path **601** of Gen X type between the Gen X port **103** of the USB host controller **101** and the Gen X port **211** (not illustrated in FIG. **4** but illustrated in FIG. **1**) of the USB device controller **209**. The connection manager **125** establishes the tunneled path **601** by enabling the Gen X adapter **109** of the USB host router **107** and the Gen X adapter **203** of the USB device router **201**. The connection manager **125** may enable the Gen X adapters **109** and **203** by programming the Path Enable (PE) control bit (e.g., writing a value of 1 for the control bit) for the Gen X adapter **109** in the control register of the USB host router **107**, and by programming the Path Enable (PE) control bit (e.g., writing a value of 1 for the control bit) for the Gen X adapter **203** in the control register of the USB device router **201**. The programming of the PE control bit of the Gen X adapter **203** may be done through a sideband channel. After the tunneled path **601** is established, the connection manager **125** informs the USB host driver **123** of the establishment of the tunneled path **601**.

Once the tunneled path **601** is established, data packets of Gen X protocol format may be transmitted on the tunneled path **601**. In some implementations, the disconnect of the tunneled path **401** and the establishment of the tunneled path **601** happens in the background without user (e.g., human user) notifications. For example, the operation system (OS) driver may suppress user notifications regarding the disconnect of the tunneled path **401** and the establishment of the tunneled path **601**.

FIGS. **4-6** show a protocol switch flow to replace a Gen T tunneled path with a Gen X tunneled path. Upon reading the discussion herein, a skilled artisan will readily be able to apply the principle disclosed herein for a protocol switch flow to replace a Gen X tunneled path with a Gen T tunneled path. In addition, the protocol switch flow can be used not only for a newly connected USB device but also for a USB device already transmitting data packets using a first type (e.g., Gen X type) of tunneled path. As discussed above, if a Gen T capable USB device is initially assigned a Gen X connection (e.g., due to lack of downstream Gen T port initially), and later if a downstream Gen T port becomes available, the principle of the protocol switching flow discussed above can be applied to upgrade the connection from Gen X connection to Gen T connection. Furthermore, if a USB hub **300** is connected between the USB device **200** and the USB host **100**, the protocol switch flow still applies and may be modified accordingly to achieve protocol switching for such a configuration. These and other variations are fully intended to be included within the scope of the present disclosure.

FIG. **7** illustrates a flow chart of a method **700** of operating a USB host system in some implementations. It should be understood that the example method shown in FIG. **7** is merely an example of many possible example methods. A skilled artisan would recognize many variations, alternatives, and modifications. For example, various steps as illustrated in FIG. **7** may be added, removed, replaced, rearranged, or repeated.

Referring to FIG. **7**, at block **710**, a first tunneled path between a first port of a USB host controller of the USB host system and a first port of a USB device controller of a USB device is disabled by a USB host driver of the USB host system, wherein the USB device is connected to the USB host system, wherein the first port of the USB host controller is coupled to a first protocol adapter of a USB host router of the USB host system, wherein the first port of the USB device controller is coupled to a first protocol adapter of a USB device router of the USB device, wherein the first tunneled path is configured to, through protocol tunneling, transmit first data packets of a first protocol format. At block **720**, a request for protocol switching is sent to a connection manager of the USB host system by the USB host driver. At block **730**, the first tunneled path is disconnected by the connection manager by disabling the first protocol adapter of the USB host router and disabling the first protocol adapter of the USB device router. At block **740**, a second tunneled path between a second port of the USB host controller and a second port of the USB device controller is established by the connection manager, wherein establishing the second tunneled path comprises enabling a second protocol adapter of the USB host router coupled to the second port of the USB host controller, and enabling a second protocol adapter of the USB device router coupled to the second port of the USB device controller.

Implementations may achieve advantages as described below. For example, the disclosed protocol switching flow may be used to disconnect a first type of tunneled path and establish a second type of tunneled path between a USB host and a USB device, while the USB device remains physically connected to the USB host. The ability to switch between different types (e.g., Gen T type or Gen X type) of tunneled paths dynamically avoids or overcomes certain incompatibility issues, and allows the USB device to achieve higher throughput when opportunity occurs. The disclosed protocol switching flow can be implemented using hardware modules (e.g., **101**, **107**) and software modules (e.g., **123**, **125**) already defined in the USB specification, and therefore, can be implemented to enhance the performance of USB systems without the need for hardware change or software change.

Examples of the present invention are summarized here. Other examples can also be understood from the entirety of the specification and the claims filed herein.

Example 1. A method of operating a Universal Serial Bus (USB) host system includes: disabling, by a USB host driver of the USB host system, a first tunneled path between a first port of a USB host controller of the USB host system and a first port of a USB device controller of a USB device, wherein the first tunneled path is configured to, through protocol tunneling, transmit first data packets of a first protocol format; sending, by the USB host driver, a request for protocol switching to a connection manager of the USB host system; disconnecting, by the connection manager, the first tunneled path; and establishing, by the connection manager, a second tunneled path between a second port of the USB host controller and a second port of the USB device controller.

Example 2. The method of Example 1, wherein the first port of the USB host controller is coupled to a first protocol adapter of a USB host router of the USB host system, wherein the first port of the USB device controller is coupled to a first protocol adapter of a USB device router of the USB device.

Example 3. The method of Example 2, further comprising, after establishing the second tunneled path, transmitting, through protocol tunneling, second data packets of a second protocol format on the second tunneled path, wherein the second protocol format is different from the first protocol format.

Example 4. The method of Example 3, wherein third data packets are configured to carry the first data packets and the second data packets in payload regions of the third data packets, wherein the third data packets are transmitted between the USB host router and the USB device router.

Example 5. The method of Example 3, further comprising: notifying, by the connection manager, the USB host driver that the first tunneled path is disconnected after disconnecting the first tunneled path and before establishing the second tunneled path; and notifying, by the connection manager, the USB host driver that the second tunneled path is established after establishing the second tunneled path and before transmitting the second data packets.

Example 6. The method of Example 3, wherein disconnecting the first tunneled path comprises: disabling the first port of the USB host controller; disabling the first protocol adapter of the USB host router; and disabling the first protocol adapter of the USB device router.

Example 7. The method of Example 6, wherein establishing the second tunneled path comprises: enabling a second protocol adapter of the USB host router coupled to the second port of the USB host controller; and enabling a second protocol adapter of the USB device router coupled to the second port of the USB device controller.

Example 8. The method of Example 7, wherein the first protocol format and the second protocol format are USB4 Gen X format and USB4 Gen T format, respectively.

Example 9. The method of Example 8, wherein the first protocol adapter of the USB host router and the first protocol adapter of the USB device router are USB4 Gen X protocol adapters, wherein the second protocol adapter of the USB host router and the second protocol adapter of the USB device router are USB4 Gen T protocol adapters.

Example 10. The method of Example 7, wherein the first protocol format and the second protocol format are USB4 Gen T format and USB4 Gen X format, respectively.

Example 11. The method of Example 10, wherein the first protocol adapter of the USB host router and the first protocol adapter of the USB device router are USB4 Gen T protocol adapters, wherein the second protocol adapter of the USB host router and the second protocol adapter of the USB device router are USB4 Gen X protocol adapters.

Example 12. A method of operating a Universal Serial Bus (USB) host system includes: transmitting first data packets of a first protocol format through a first tunneled path between a first port of a USB host controller of the USB host system and a first port of a USB device controller of a USB device, wherein the USB device is connected to the USB host system; stopping, by a USB host driver of the USB host system, transmission of the first data packets by disabling the first port of the USB host controller; sending, by the USB host driver, a request for protocol switching to a connection manager of the USB host system; tearing down, by the connection manager, the first tunneled path; establishing, by the connection manager, a second tunneled path between a second port of the USB host controller and a second port of the USB device controller; and transmitting second data packets of a second protocol format through the second tunneled path, wherein the second protocol format is different from the first protocol format.

Example 13. The method of Example 12, wherein stopping transmission of the first data packets comprises: programming, by the USB host driver, a control bit in a control register of the USB host controller to disable the first port of the USB host controller.

Example 14. The method of Example 12, wherein tearing down the first tunneled path comprises: disabling, by the connection manager, a first protocol adapter of a USB host router of the USB host system connected to the first port of the USB host controller; and disabling, by the connection manager, a first protocol adapter of a USB device router of the USB device connected to the first port of the USB device controller.

Example 15. The method of Example 14, wherein establishing the second tunneled path comprises: enabling, by the connection manager, a second protocol adapter of the USB host router connected to the second port of the USB host controller; and enabling, by the connection manager, a second protocol adapter of the USB device router connected to the second port of the USB device controller.

Example 16. In an implementation, a Universal Serial Bus (USB) host system includes: a USB host controller having a first port and a second port, wherein the first port and the second port of the USB host controller are configured to transmit data packets in a first protocol format and a second protocol format, respectively; a USB host driver that drives the USB host controller; a USB connector for connection with a USB device; a USB host router having a first protocol adapter and a second protocol adapter, wherein the USB host router is configured to transmit data packets in a third protocol format via the USB connector, wherein the first protocol adapter and the second protocol adapter of the USB host router are coupled to the first port and the second port of the USB host controller, respectively; and a connection manager coupled to the USB host driver and the USB host router, wherein the USB host system is configured to: disable, by the USB host driver, a first tunneled path between the first port of the USB host controller and a first port of a USB device controller of the USB device; send, by the USB host driver, a request for protocol switching to the connection manager; tear down, by the connection manager, the first tunneled path; and establish, by the connection manager, a second tunneled path between the second port of the USB host controller and a second port of the USB device controller.

Example 17. The USB host system of Example 16, wherein the USB host controller and the USB host router belong to a hardware domain of the USB host system, and wherein the USB driver and the connection manager belong to a software domain of the USB host system.

Example 18. The USB host system of Example 16, wherein the USB host system is configured to disable the first tunneled path by disabling the first port of the USB host controller.

Example 19. The USB host system of Example 18, wherein the first port of the USB device controller is coupled to a first protocol adapter of a USB device router of the USB device, wherein the USB host system is configured to tear down the first tunneled path by disabling the first protocol adapter of the USB host router and by disabling the first protocol adapter of the USB device router, wherein the USB host system is configured to establish the second tunneled path by enabling the second protocol adapter of the USB host router and by enabling a second protocol adapter of the USB device router coupled to the second port of the USB device controller.

Example 20. The USB host system of Example 16, wherein the USB host system is further configured to: after establishing the second tunneled path, transmit data packets between the USB host controller and the USB device controller via the second tunneled path.

While this invention has been described with reference to illustrative examples, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative examples, as well as other examples of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or examples.

What is claimed is:

1. A method of operating a Universal Serial Bus (USB) host system, the method comprising:

disabling, by a USB host driver of the USB host system, a first tunneled path between a first port of a USB host controller of the USB host system and a first port of a USB device controller of a USB device, wherein the first tunneled path is configured to, through protocol tunneling, transmit first data packets of a first protocol format;

sending, by the USB host driver, a request for protocol switching to a connection manager of the USB host system;

disconnecting, by the connection manager, the first tunneled path; and

establishing, by the connection manager, a second tunneled path between a second port of the USB host controller and a second port of the USB device controller.

2. The method of claim **1**, wherein the first port of the USB host controller is coupled to a first protocol adapter of a USB host router of the USB host system, wherein the first port of the USB device controller is coupled to a first protocol adapter of a USB device router of the USB device.

3. The method of claim **2**, further comprising, after establishing the second tunneled path, transmitting, through protocol tunneling, second data packets of a second protocol format on the second tunneled path, wherein the second protocol format is different from the first protocol format.

4. The method of claim **3**, wherein third data packets are configured to carry the first data packets and the second data packets in payload regions of the third data packets, wherein the third data packets are transmitted between the USB host router and the USB device router.

5. The method of claim **3**, further comprising:

notifying, by the connection manager, the USB host driver that the first tunneled path is disconnected after disconnecting the first tunneled path and before establishing the second tunneled path; and

notifying, by the connection manager, the USB host driver that the second tunneled path is established after establishing the second tunneled path and before transmitting the second data packets.

6. The method of claim **3**, wherein disconnecting the first tunneled path comprises:

disabling the first port of the USB host controller;

disabling the first protocol adapter of the USB host router; and

disabling the first protocol adapter of the USB device router.

7. The method of claim **6**, wherein establishing the second tunneled path comprises:

enabling a second protocol adapter of the USB host router coupled to the second port of the USB host controller; and

enabling a second protocol adapter of the USB device router coupled to the second port of the USB device controller.

8. The method of claim **7**, wherein the first protocol format and the second protocol format are USB4 Gen X format and USB4 Gen T format, respectively.

9. The method of claim **8**, wherein the first protocol adapter of the USB host router and the first protocol adapter of the USB device router are USB4 Gen X protocol adapters, wherein the second protocol adapter of the USB host router and the second protocol adapter of the USB device router are USB4 Gen T protocol adapters.

10. The method of claim **7**, wherein the first protocol format and the second protocol format are USB4 Gen T format and USB4 Gen X format, respectively.

11. The method of claim **10**, wherein the first protocol adapter of the USB host router and the first protocol adapter of the USB device router are USB4 Gen T protocol adapters, wherein the second protocol adapter of the USB host router and the second protocol adapter of the USB device router are USB4 Gen X protocol adapters.

12. A method of operating a Universal Serial Bus (USB) host system, the method comprising:

transmitting first data packets of a first protocol format through a first tunneled path between a first port of a USB host controller of the USB host system and a first port of a USB device controller of a USB device, wherein the USB device is connected to the USB host system;

stopping, by a USB host driver of the USB host system, transmission of the first data packets by disabling the first port of the USB host controller;

sending, by the USB host driver, a request for protocol switching to a connection manager of the USB host system;

tearing down, by the connection manager, the first tunneled path;

establishing, by the connection manager, a second tunneled path between a second port of the USB host controller and a second port of the USB device controller; and

transmitting second data packets of a second protocol format through the second tunneled path, wherein the second protocol format is different from the first protocol format.

13. The method of claim **12**, wherein stopping transmission of the first data packets comprises:

programming, by the USB host driver, a control bit in a control register of the USB host controller to disable the first port of the USB host controller.

14. The method of claim **12**, wherein tearing down the first tunneled path comprises:

disabling, by the connection manager, a first protocol adapter of a USB host router of the USB host system connected to the first port of the USB host controller; and

disabling, by the connection manager, a first protocol adapter of a USB device router of the USB device connected to the first port of the USB device controller.

15. The method of claim **14**, wherein establishing the second tunneled path comprises:

enabling, by the connection manager, a second protocol adapter of the USB host router connected to the second port of the USB host controller; and

enabling, by the connection manager, a second protocol adapter of the USB device router connected to the second port of the USB device controller.

16. A Universal Serial Bus (USB) host system comprising:

a USB host controller having a first port and a second port, wherein the first port and the second port of the USB host controller are configured to transmit data packets in a first protocol format and a second protocol format, respectively;

a USB host driver that drives the USB host controller;

a USB connector for connection with a USB device;

a USB host router having a first protocol adapter and a second protocol adapter, wherein the USB host router is configured to transmit data packets in a third protocol format via the USB connector, wherein the first protocol adapter and the second protocol adapter of the USB host router are coupled to the first port and the second port of the USB host controller, respectively; and

a connection manager coupled to the USB host driver and the USB host router,

wherein the USB host system is configured to:

disable, by the USB host driver, a first tunneled path between the first port of the USB host controller and a first port of a USB device controller of the USB device;

send, by the USB host driver, a request for protocol switching to the connection manager;

tear down, by the connection manager, the first tunneled path; and

establish, by the connection manager, a second tunneled path between the second port of the USB host controller and a second port of the USB device controller.

17. The USB host system of claim **16**, wherein the USB host controller and the USB host router belong to a hardware domain of the USB host system, and wherein the USB driver and the connection manager belong to a software domain of the USB host system.

18. The USB host system of claim **16**, wherein the USB host system is configured to disable the first tunneled path by disabling the first port of the USB host controller.

19. The USB host system of claim **18**, wherein the first port of the USB device controller is coupled to a first protocol adapter of a USB device router of the USB device, wherein the USB host system is configured to tear down the first tunneled path by disabling the first protocol adapter of the USB host router and by disabling the first protocol adapter of the USB device router, wherein the USB host system is configured to establish the second tunneled path by enabling the second protocol adapter of the USB host router and by enabling a second protocol adapter of the USB device router coupled to the second port of the USB device controller.

20. The USB host system of claim **16**, wherein the USB host system is further configured to:

after establishing the second tunneled path, transmit data packets between the USB host controller and the USB device controller via the second tunneled path.

* * * * *